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RESEARCH ARTICLE

Brazilian Sign Language Recognition Using Deep Learning Based on Fast Fourier Transform and Kinematic Features

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ABSTRACT Robust sign language recognition necessitates feature representations capable of jointly modeling the spatial articulations of the hands, arms, and body; the temporal dynamics of sequential gestures; and frequency-domain patterns that characterize fluency, rhythm, and coarticulation in signing. Unlike general human action recognition, sign language understanding requires fine-grained discrimination of subtle handshapes, orientations, and facial expressions, all of which are tightly coupled with temporal dependencies across frames. This paper proposes a comprehensive feature extraction for motion capture sequences, augmenting raw joint positions with kinematic features, such as velocities and accelerations, applied to Brazilian Sign Language recognition. To further capture characteristic motion frequencies, we introduce an adaptive sliding-window Fast Fourier Transform (FFT) analysis, extracting dominant frequency magnitudes and phase information. The resulting improved feature set integrates geometric, temporal, and spectral information, enabling improved performance in sign recognition. Evaluated on the MINDS-Libras dataset, the proposed method achieved 92% accuracy and 92% F1-score using 10-fold cross-validation, outperforming baseline LSTM (84%), BiLSTM (89%), and CNN (89%) models, as well as surpassing the state-of-the-art methods. Experimental results demonstrate the effectiveness of the proposed multi-modal representation compared to baseline approaches relying solely on positional data. Real-world experiments in both indoor and outdoor settings demonstrated a performance with average classification probabilities of 99.32% and 98.20%, respectively, with feature extraction processing at 0.148 seconds per frame and model inference at 0.198 seconds.

INDEX TERMS Deep learning, fast fourier transform, Libras, sign language recognition.

I. INTRODUCTION

The World Federation of the Deaf (WFD) states that deafness creates communication barriers that often prevent people who are deaf from fully taking part in society [1]. In Brazil, the deaf community uses Brazilian Sign Language (Libras) as their first language. Libras is a natural visual-spatial language used by the Brazilian deaf community [2]. Like other sign languages around the world, Libras has a full linguistic structure, including its phonology, morphology, syntax, and semantics [3], [4]. It combines hand gestures with facial and body expressions, allowing people to communicate ideas, emotions, and concepts smoothly [5]. Even though Libras was officially recognized as a national language in 2002 by Law No. 10,436, communication between hearing and deaf people still faces many challenges, and deaf individuals remain part of a linguistic and cultural minority [1], [6].

The Brazilian Institute of Geography and Statistics (IBGE) reports that approximately 5% of Brazil's population, or around 10 million people, experience some level of hearing

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loss. Nearly three million of them are completely deaf [7]. Although there are no precise statistics on how many people know Libras, it is estimated that only a small fraction of Brazil's population of over 200 million does.

The social inclusion of deaf people largely depends on overcoming communication barriers. However, many people in Brazil are not fluent in Libras, and there is a shortage of interpreters, which makes inclusion difficult [8], [9]. In this context, digital tools like VLibras [2] and SpreadTheSign [10] play an important role by providing access to Libras vocabulary, supporting self-learning, and offering exposure to different sign languages. These tools help spread deaf culture and improve communication accessibility. Additionally, artificial intelligence (AI)–based technologies are emerging as a promising solution, enabling the development of automatic translation tools between Libras and spoken or written Portuguese [4], [11], [12].

AI models, such as Convolutional Neural Networks (CNNs), are widely used in computer vision for tasks such as image classification, object detection, and segmentation. Over time, several architectures have been proposed, including LeNet-5 [13], AlexNet [14], VGG [15], Inception [16], ResNet [17], MobileNet [18], EfficientNet [19], and DenseNet [20]. Each aims to improve depth, efficiency, or training stability, and the choice depends on task requirements and computational resources.

In sign language recognition (SLR), CNNs are widely adopted for their ability to extract spatial features from images and video frames [21], [22], [23], [24]. They identify hand shapes, gestures, and facial expressions, with deep residual models like ResNet excelling on large datasets [25]. For real-time or mobile scenarios, lightweight networks, such as MobileNet and EfficientNet, are preferred [26], [27]. To capture motion in dynamic gestures, CNNs are often combined with Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM), where CNNs extract spatial features and RNNs model temporal dependencies [28], [29].

One of the most significant barriers to effective Sign Language Recognition (SLR) systems, particularly for Libras, is the scarcity of diverse datasets [30]. This limits the ability of AI models, like CNNs, to generalize well across real-world variations such as different signers, backgrounds, lighting, and co-articulated gestures. Several Libras datasets exist, each with specific limitations as summarized in Table 1. Most datasets are either limited to static fingerspelling images [31], [32], not publicly available [33], or have restricted vocabulary [34]. The MINDS-Libras dataset [34], [35], containing 1,200 video sequences of 20 signs performed by 12 signers with five repetitions each, was selected for this study due to its public accessibility and provision of dynamic sign videos, enabling the extraction of both spatial and temporal features. Unlike static image datasets, MINDS-Libras comprises 20 distinct signs, each performed multiple times by 12 signers, enabling the capture of both spatial and temporal features essential for dynamic sign recognition.

TABLE 1. Summary of Libras datasets and their challenges.

Dataset	Type	Size	Challenges
Alphabet [31]	Static RGB images	3,000 images (15 signs)	Limited signs; homogeneous setup
UFPB Fingerspelling [32]	RGB images	224,000 (full alphabet)	Reduced accuracy on unseen signers
LIBRAS-UFOP [33]	RGB-D + Skeleton (video)	56 signs; dynamic minimal pairs	Challenging to distinguish similar signs
MINDS-Libras [34]	RGB video	20 signs, 12 signers	Limited vocabulary set
Facial Expression [36], [37]	RGB-D or skeleton	Small scale	Narrow expression and signer diversity

Developing an effective Libras SLR system involves several key challenges. First, temporal variability makes consistent modeling difficult, as signers perform the same signs with different speeds and rhythms, further complicated by co-articulation effects that blur sign boundaries. Second, inter-signer diversity introduces large appearance and motion differences due to variations in anatomy and signing style, demanding strong generalization. Third, many signs are visually similar, differing only in subtle kinematic or spatial cues, requiring fine-grained feature representations. Fourth, effective feature design must jointly encode spatial, temporal, and spectral information while remaining computationally efficient. Finally, real-time performance is essential for practical use, yet landmark detection, kinematic computation, and spectral analysis can be computationally demanding, especially on limited hardware.

To address these challenges, this work advances the state-of-the-art in Libras recognition through three key innovations that distinguish it from existing approaches. First, while most prior work in SLR relies exclusively on spatial features from hand landmarks [38], [39], [40] or combines spatial features with basic temporal modeling through LSTMs [29], [41], the proposed approach introduces a multi-modal feature fusion framework that integrates three complementary information sources: (i) geometric hand landmark positions from MediaPipe for spatial configuration, (ii) first and second-order kinematic descriptors (velocity and acceleration) to capture motion dynamics, and (iii) frequency-domain representations via adaptive sliding-window FFT to encode rhythmic and temporal patterns. Second, we demonstrate through comprehensive experimental validation that this multi-modal approach yields substantial performance improvements over existing methods. More importantly, feature importance analysis reveals that spectral features contribute to disambiguating similar signs, a finding that validates our hypothesis that motion dynamics and frequency characteristics provide discriminative information not captured by spatial features alone. These results establish a new benchmark for Libras recognition and provide a replicable framework that can be adapted to other sign languages. Therefore, the main contributions of this work are as follows:

i) Propose a multi-modal feature extraction combining positional, kinematic (velocity and acceleration), and spectral (FFT-based) information from motion capture sequences;
ii) Introduce an adaptive sliding-window FFT approach to extract dominant motion frequencies and phase components;
iii) Demonstrate, through experimental evaluation, that the enriched feature set improves performance in downstream tasks such as gesture recognition and movement classification compared to baselines using only positional data.

The paper is organized as follows: Section II reviews related work in sign language recognition techniques. Section III presents the methodology, including the dataset description, the proposed multi-modal feature extraction, and the formulation of the proposed model for sign language recognition. Section IV describes the experimental setup, evaluation metrics, and discusses the experimental results obtained from both simulation and real-world tests. Finally, Section V concludes the paper and outlines directions for future research.

II. RELATED WORKS

Table 2 summarizes recent SLR models, highlighting their approaches, accuracies, datasets, applications, and target sign languages. Deep learning models, particularly CNNs, RNNs (LSTM), hybrid CNN-RNNs, and attention-based architectures, have been extensively applied to SLR. Architecture selection depends on whether the task involves static images or dynamic video sequences, and on the complexity and variability of the target sign language. In the work [43], the authors presented a comprehensive survey of vision-based sign language recognition using deep learning. They covered architectures including CNN, LSTM, 3D-CNN, Restricted Boltzmann Machine (RBM), Variational Autoencoders (VAE), and hybrid models, discussing limitations such as feature fusion challenges, limited multi-person recognition, and the exclusion of sensor-based methods.

Many recent studies have focused on the application of CNNs and LSTMs for sign language recognition. Wadhawan and Kumar [38] developed an Indian Sign Language (ISL) recognition system based on a CNN architecture. Evaluated on colored and grayscale datasets, the model achieved training accuracy of 99.90% and validation accuracy of 98.70% with the Stochastic Gradient Descent (SGD) optimizer, while Adaptive Moment Estimation (Adam) yielded 99.17% and 98.80%, respectively. Also using ISL, Sharma et al. [39] conducted a comparative analysis of three deep learning architectures for ISL gesture recognition: a pre-trained VGG16 model, a natural language–based output network, and a hierarchical network. Using approximately 149,568 gesture images, the fine-tuned VGG16 achieved 94.52% accuracy, while the hierarchical model reached 98.52% for one-hand gestures and 97% for two-hand gestures. The study noted the need for better handling of similar gestures, improved sentence generation, and faster inference for long sequences.

Rastgoo et al. [41] proposed a deep cascaded framework for Persian sign recognition, integrating SSD, CNN, and LSTM components with VGG16 and ResNet-50 features. On their proprietary dataset, the system reached 98.42% accuracy and 83.45% on the isoGD dataset. As [38] and [39], Sharma and Singh [42] proposed G-CNN, a robust architecture for ISL recognition, tested on a 43-class dataset. Compared with VGG-11 and VGG-16, G-CNN achieved 94.83% for Category-1 and 99.96% for Category-2. Future work focused on real-time optimization and reducing recognition errors.

Using American Sign Language (ASL), Kasapbasi et al. [40] designed a CNN model for ASL recognition, introducing the large-scale ASLA dataset. Their system achieved 99.41%, 99.48%, and 99.38% accuracy on three datasets. Natarajan et al. [29] developed an end-to-end framework combining recognition, translation, and video generation. The recognition component employed a CNN and Bi-LSTM architecture with MediaPipe for gesture extraction, achieving over 95% accuracy on multilingual datasets including ISL-CSLTR and RWTH-PHOENIX-Weather 2014T. Video generation was powered by an NMT–Dynamic GAN hybrid, producing realistic sign videos.

Attia et al. [44] optimized YOLOv5x with SE and CBAM attention modules for Bangla sign recognition, achieving 98.9% precision, 96.5% recall, and 99.2% mAP on MU HandImages ASL and OkkhorNama datasets. Building on alternative feature representations, Bayrak et al. [46] introduced CVDNN, leveraging the magnitude and phase of Complex Zernike Moments for ASL recognition. On the MNIST ASL dataset, it achieved 89.01%, and on the Massey University Dataset, 98.67% (holdout) and 81.22% (LOSO). Rakshit et al. [47] developed two CNN models for grayscale ASL alphabet detection, achieving 98.19% accuracy.

Rezende et al. [35] introduced the MINDS-Libras dataset, containing over 1,000 RGB and depth videos of 20 Brazilian Sign Language signs from 12 participants. Baseline CNN experiments achieved 93.3% accuracy. De Castro et al. [12] proposed a 3D CNN method processing only RGB input, indirectly estimating pose and depth. The model reached 96% accuracy in 10-fold cross-validation but dropped to around 57% with leave-one-person-out testing, indicating overfitting to signer-specific features.

Arcanjo et al. [45] explored Fast Dynamic Time Warping (FastDTW) for Libras video recognition. Their method calculates angular features from landmarks to perform sign recognition without deep learning techniques, achieving 96% accuracy on MINDS-Libras and 90% on INCLUDE-50. However, the approach requires improvements in addressing signs with similar hand configurations, such as "Bathroom" and "Frog". In the manuscript [4], the authors applied several machine learning models, including Extra Trees, Random Forest, and K-Nearest Neighbors, achieving accuracies of 98%, 97%, and 94%, respectively, on the MINDS-Libras

TABLE 2. Summary of reviewed sign language recognition models.

Reference	Approach	Accuracy	Dataset(s)	Application	Sign Language
[41]	Deep cascaded model (SSD + CNN + LSTM) with VGG16 and ResNet50 features	98.42% (own), 83.45% (isoGD)	Own dataset, isoGD	Hand sign recognition; feature fusion; proposal to expand dictionary and sample diversity	Persian
[39]	VGG16 (TL / fine-tuning), natural language output network, hierarchical network	53% (TL), 94.52% (fine-tuned VGG16), 98.52% (1-hand), 97% (2-hand)	∼149k ISL images (black bg)	ISL sentence generation; improve similar gesture classification	Indian
[38]	CNN (SGD / Adam)	98.70% (SGD); 98.80% (Adam)	Colored and grayscale ISL datasets	Optimize for large datasets	Indian
[35]	CNN	93.3%	MINDS-Libras	Libras database creation; RGB + depth	Libras
[42]	G-CNN	94.83% (Cat-1), 99.96% (Cat-2)	43-class ISL dataset	Robust ISL recognition; compared with VGG-11/16; future real-time optimization	Indian
[43]	Survey of CNN, LSTM, 3DCNN, RBM, VAE, hybrid models	–	Multiple	Review of vision-based SLR models; focus on DL, not sensor-based	Multiple
[40]	CNN (ASLA dataset)	99.41%, 99.48%, 99.38%	ASLA + 2 other datasets	Predict letters/words; potential speech/robot control	ASL
[29]	End-to-end (CNN + Bi-LSTM + MediaPipe) + NMT + Dynamic GAN	>95% (recognition), BLEU 38.56, FID 3.46	ISL-CSLTR, RWTH-PHOENIX-Weather 2014T, others	Recognition, translation, and video generation in one pipeline	Indian, German, Multilingual
[12]	3D CNN using RGB input (pose/depth estimated)	96% (10-fold), ∼57% (leave-one-person-out)	MINDS-Libras, LIBRAS-UFOP, INCLUDE-50, KSL	No depth/gloves; evaluates generalization limits; improve depth map gen.	Libras, KSL
[44]	SE-YOLOv5x, CBAM-YOLOv5x, CBAMELU-YOLOv5x	Prec. 98.9%, Rec. 96.5%, mAP 99.2% (best)	MU HandImages ASL, OkkhorNama	Object detection + attention; Bangla SLR	Bangla, American
[45]	FastDTW	–	MINDS-Libras	Temporal alignment for Libras gesture recognition	Libras
[46]	CVDNN (CZM-based DNN)	89.01% (MNIST), 98.67% (holdout), 81.22% (LOSO)	MNIST ASL (24 classes), Massey University Dataset (36 classes)	Uses magnitude + phase of CZMs; improve robustness	American
[47]	Two CNN models	98.19%	ASL alphabet dataset (grayscale)	Alphabet detection; investigate label-specific performance	American
[48]	Hybrid GCN + MHSA + CNN (GmTC)	99.33%, 100% (KSL), 93.50%, 96.88% (BSL)	KSL, JSL, BSL, ASL, others	Multi-culture SLR; high accuracy across datasets; real-time efficiency focus	Korean, Japanese, British, American, Bangla

dataset. However, they considered only hand position, disregarding facial expressions, which are crucial for recognizing certain signs.

Finally, Miah et al. [48] presented Graph meets with attention and CNN (GmTC), an end-to-end multi-lingual recognition model combining Graph Convolutional Network (GCN), multi-head self-attention, and CNN. Evaluated on datasets from Korean Sign Language (KSL), Japanese Sign Language (JSL), Bangla Sign Language (BSL), ASL, and others, it achieved up to 100% on KSL and 96.88% on BSL, outperforming prior works. Remaining challenges include improving generalization, handling dynamic gestures, and reducing computational cost.

Recent trends in SLR have increasingly explored transformer-based architectures, which leverage self-attention mechanisms to capture long-range dependencies in sequential data without the vanishing gradient limitations of RNNs. Transformers have shown promising results in continuous sign language translation tasks, particularly for modeling complex temporal relationships and handling variable-length sequences [49], [50], [51]. Additionally, self-supervised learning approaches have emerged as effective strategies to address the scarcity of labeled sign language data [52], [53]. Multi-modal fusion strategies represent another research direction: early fusion concatenates features from different modalities (e.g., RGB, depth, optical flow, landmarks) before processing, enabling joint learning but requiring careful normalization; late fusion processes each modality independently through specialized networks and combines predictions at the decision level, offering modularity but potentially missing cross-modal interactions [54], [55], [56].

Despite significant advancements in SLR across various languages, research on Libras remains comparatively limited and faces important challenges. While many existing models achieve high accuracy for American, Indian, or other sign languages, Libras recognition systems often struggle with limited and less diverse datasets, overfitting to specific signers, and reduced generalization to real-world scenarios. Therefore, there is a need for improved methods that can accurately detect and interpret Libras signs across different signers, lighting conditions, and backgrounds, enabling more reliable and inclusive applications for the Brazilian deaf community.

III. METHODOLOGY

This section presents the methodology, including dataset description, multi-modal feature extraction, and the proposed model for Brazilian Sign Language recognition.

A. DATASET

The MINDS-Libras dataset [34], [35] was used for this study. Fig. 1 shows sample frames representing the signs "Happen", "Yellow", "Son", and "Five".

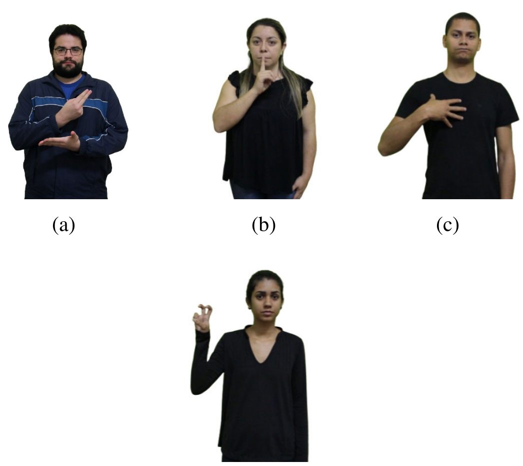


FIGURE 1. Sample frames from the dataset MINDS-Libras illustrating four different classes: (a) "Happen", (b) "Yellow", (c) "Son", and (d) "Five".

The dataset contains 1,200 video sequences across 20 classes, representing different Libras signs. Each sign was recorded 5 times by 12 signers, providing diverse examples for training and evaluation. Data was collected using a Chroma Key background, and signers included men and women with basic to advanced Libras knowledge. Videos are 1920×1080 MP4 files recorded at 30 fps [34], [35].

Since videos are frame sequences, temporal dynamics must be preserved. Each frame provides critical information about the signer's pose and hand movements. We implemented a feature extraction procedure that processes every frame individually, preserving both spatial and temporal information. Hand and upper limb coordinates are extracted from each frame. For symmetric signs, the procedure is applied to mirrored frames, enriching the dataset with both right- and left-handed perspectives. Metadata records the video identifier, frame index, and mirror status. Finally, processed frames are grouped by source video to form time series representations.

Fig. 2 illustrates the procedure applied to the input video frames to extract the features. The method is applied to both original and mirrored frames, ensuring that the dataset includes signs from both right- and left-handed signers.

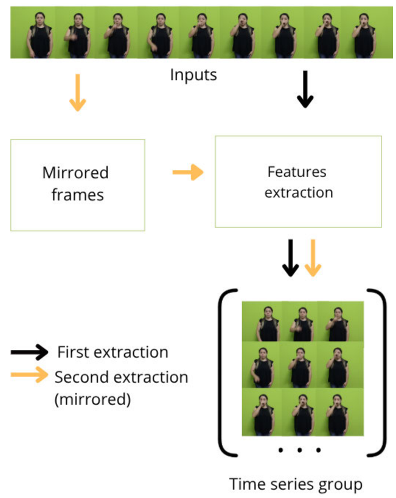


FIGURE 2. Procedure applied in the input video frames to extract the features. The procedure is applied to both original and mirrored frames, ensuring that the dataset includes signs from both right and left-handed signers.

The following sections describe the feature extraction process in detail, which includes landmark detection, kinematic features, and FFT-based features.

B. FEATURES EXTRACTION

Let the input image be $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$. A CNN-based Single Shot Detector (SSD) predicts a set of anchor boxes. Let the CNN feature map be:

$$\mathbf{F} = \mathrm{CNN}_{\phi}(\mathbf{I}), \quad \mathbf{F} \in \mathbb{R}^{H' \times W' \times C}, \tag{1}$$

where H', W' are reduced spatial dimensions and C channels.

An anchor box $(x^{\text{pred}}_{\text{center}}, y^{\text{pred}}_{\text{center}}, w^{\text{pred}}, h^{\text{pred}})$ is predicted from anchor (x_a, y_a, w_a, h_a) by

$$x^{\text{pred}}_{\text{center}} = \Delta x\, w_a + x_a, \tag{2}$$
$$y^{\text{pred}}_{\text{center}} = \Delta y\, h_a + y_a, \tag{3}$$
$$w^{\text{pred}} = w_a e^{\Delta w}, \quad h^{\text{pred}} = h_a e^{\Delta h}, \tag{4}$$

where $(\Delta x, \Delta y, \Delta w, \Delta h, s)$ are predicted by the CNN and s is the confidence score. Keep boxes where $s > 0.5$.

To remove overlapping boxes B, the Intersection over Union (IoU) for two boxes B_i, B_j is calculated as:

$$\mathrm{IoU}(B_i, B_j) = \frac{\mathrm{area}(B_i \cap B_j)}{\mathrm{area}(B_i \cup B_j)}. \tag{5}$$

The box with the highest score is selected, and boxes with IoU $>$ threshold are discarded. Non-maximum suppression selects the Region of Interest (ROI):

$$\mathrm{ROI} = (x_{\min}, y_{\min}, x_{\max}, y_{\max}). \tag{6}$$

The ROI coordinates are normalized to [0, 1]:

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad \tilde{y} = \frac{y - y_{\min}}{y_{\max} - y_{\min}}. \tag{7}$$

The ROI is cropped and resized to a fixed size for the landmark model:

$$\mathbf{I}_{\mathrm{ROI}} = \mathrm{Resize}(\mathrm{crop}(\mathbf{I}, \mathrm{ROI})). \tag{8}$$

The landmark network f_θ predicts N 3D landmarks:

$$f_\theta : \mathbf{I}_{\mathrm{ROI}} \mapsto \mathbf{L} = \{(\tilde{x}_i, \tilde{y}_i, z_i)\}_{i=1}^{N}, \tag{9}$$

$x_i, y_i \in [0, 1]$ (normalized coordinates in ROI), $z_i \in \mathbb{R}$ (relative depth). Map normalized ROI coordinates to original image coordinates:

$$X_i = \tilde{x}_i(x_{\max} - x_{\min}) + x_{\min}, \tag{10}$$

$$Y_i = \tilde{y}_i(y_{\max} - y_{\min}) + y_{\min}, \tag{11}$$

$$Z_i = s_z z_i. \tag{12}$$

where s is an optional scale factor for depth. So the final 3D landmark in image space is:

$$\mathbf{P}_i = (X_i, Y_i, Z_i). \tag{13}$$

To reduce jitter, exponential smoothing over time t is applied:

$$\hat{\mathbf{P}}_i^{(t)} = \alpha \mathbf{P}_i^{(t)} + (1 - \alpha)\mathbf{P}_i^{(t-1)}, \quad \alpha \in [0, 1]. \tag{14}$$

Fig. 3 illustrates hands landmarks extracted using the procedure described by equations (1) to (14). The image shows the 21 landmarks detected on a hand. These landmarks are extracted from the hand ROI and represent key points such as the wrist, fingertips, and knuckles. The lines connecting these points represent the relationships between them, forming a skeletal structure of the hand. The coordinates of these landmarks are normalized to fit within the ROI, and their relative depth is also captured. The smoothing process helps to reduce noise in the landmark positions over time.

Fig. 4 depicts pose landmarks extracted using the same procedure as the hand landmarks. However, in this case, the landmarks correspond to key points on the human body, such as the shoulders, elbows, wrists, hips, knees, and ankles.

The lines connecting these points represent the skeletal structure of the body. The pose landmarks are extracted from the body ROI, and their coordinates are normalized to fit within the ROI, with relative depth captured as well. The smoothing process is applied to reduce noise in the landmark positions over time. These pose landmarks are crucial for understanding the spatial configuration of the body during sign language gestures, as they provide information about the position and facial expression of the signer, which can significantly affect the meaning of the signs.

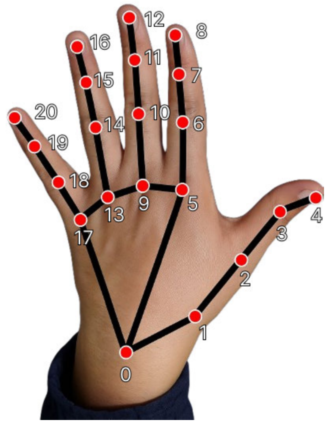


FIGURE 3. Hands landmarks with key points: spatial configuration of the hand during sign language gestures.

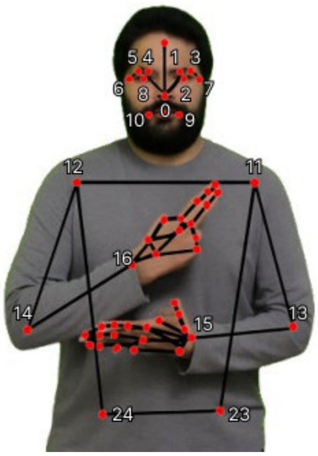


FIGURE 4. Pose landmarks with key points: spatial configuration of the body during sign "Happen" gestures.

All the procedures described from equations (1) to (14) are implemented by the framework Mediapipe in the modules Hands and Pose. The main difference between them is the number of landmarks that are extracted and the ROI that is used as input to the landmark model. The Mediapipe Hands extracts 21 landmarks per hand, while the Mediapipe Pose extracts 33 landmarks for the whole body and face [57]. In this work, only the first 17 pose landmarks are used, which correspond to the upper limbs.

C. THE PROPOSED APPROACH

In this section, we describe how positional features are enhanced with kinematic and spectral descriptors. Section III-C1 details velocity and acceleration computation to capture motion dynamics. Section III-C1b explains the adaptive sliding-window FFT used to extract dominant frequency magnitudes and normalized phase information.

1) KINEMATIC AND SPECTRAL DESCRIPTORS

The proposed approach enhances the raw positional features extracted from the sign language dataset by incorporating kinematic and spectral descriptors. The kinematic descriptors include joint velocities and accelerations, which capture the dynamic aspects of signing, such as movement speed and direction. The spectral descriptors are derived from an adaptive sliding-window FFT, which analyzes the frequency content of the motion data to identify dominant frequencies and rhythmic patterns. Therefore, to define the kinematic and spectral descriptors, let $\mathbf{p}_t^{(n)} \in \mathbb{R}^{L_p}$ be the concatenated 3D positions (including depth) at timestep t of sequence n, with

$$L_p = 3 \cdot (21 \cdot 2 + 17) = 177, \tag{15}$$

where 21 is the number of hand landmarks, 2 accounts for both left and right hands, and 17 is the number of pose landmarks used (the first 17 pose landmarks correspond to the upper limbs).

a: VELOCITY AND ACCELERATION

For $t = 1, \ldots, T_n - 1$, we computed:

$$\mathbf{v}_t^{(n)} = \mathbf{p}_{t+1}^{(n)} - \mathbf{p}_t^{(n)}, \qquad \mathbf{v}_{T_n}^{(n)} = \mathbf{0}_{L_p}, \tag{16}$$

and for $t = 1, \ldots, T_n - 2$:

$$\mathbf{a}_t^{(n)} = \mathbf{v}_{t+1}^{(n)} - \mathbf{v}_t^{(n)}, \qquad \mathbf{a}_{T_n-1}^{(n)} = \mathbf{a}_{T_n}^{(n)} = \mathbf{0}_{L_p}. \tag{17}$$

b: FAST FOURIER TRANSFORM

To calculate the FFT features, consider a sequence n of length T_n. Define a window function $w : \{1, \ldots, T_n\} \to \mathbb{R}$ and windowed input:

$$\tilde{\mathbf{p}}_t^{(n)} = w(t)\mathbf{p}_t^{(n)}. \tag{18}$$

Assuming a sampling rate f_s, the FFT frequency bins are

$$f_k = \frac{kf_s}{T_n}, \quad k = 0, \ldots, \lfloor T_n/2 \rfloor, \tag{19}$$

with frequency mask

$$\mathcal{F} = \{k \mid f_{\min} \le f_k \le f_{\max}\}. \tag{20}$$

The FFT coefficients for feature i are defined as:

$$X_i^{(n)}[k] = \sum_{t=1}^{T_n} \tilde{\mathbf{p}}_{t,i}^{(n)} e^{-j2\pi(t-1)k/T_n}, \quad k \in \mathcal{F}. \tag{21}$$

Magnitude and phase features are defined as:

$$M_i^{(n)}[k] = |X_i^{(n)}[k]|, \tag{22}$$

and

$$\Phi_i^{(n)}[k] = \frac{\arg(X_i^{(n)}[k])}{\pi} \in [-1, 1]. \tag{23}$$

The top n_{coeffs} frequency indices are selected based on the total magnitude:

$$\mathcal{K} = \arg \max_{\substack{\mathcal{S} \subseteq \mathcal{F} \\ |\mathcal{S}| = n_{\text{coeffs}}}} \sum_{i=1}^{L_p} \sum_{k \in \mathcal{S}} M_i^{(n)}[k]. \tag{24}$$

From the FFT feature vectors, for each sequence n and timestep t, the magnitude and phase features are calculated as:

$$\mathbf{m}^{(n)} = \{M_i^{(n)}[k]\}_{k \in \mathcal{K}, i=1}^{L_p} \tag{25}$$

and

$$\boldsymbol{\phi}^{(n)} = \{\Phi_i^{(n)}[k]\}_{k \in \mathcal{K}, i=1}^{L_p}. \tag{26}$$

c: ADAPTIVE SLIDING-WINDOW FFT

An adaptive sliding-window FFT is applied to the sequence of 3D positions $\mathbf{p}_t^{(n)}$ to extract frequency-based features. The procedure is as follows:

Consider a window size W that is adaptively determined based on the length of the sequence T_n:

$$W = \min\left(\max\left(\left\lfloor \frac{T_n}{\alpha} \right\rfloor, W_{\min}\right), W_{\max}\right), \tag{27}$$

where $\alpha > 0$, $W_{\min}$ is the minimum and $W_{\max}$ is the maximum allowed window size.

The window size formula (27) ensures that the FFT analysis window scales proportionally with sequence duration while maintaining local temporal resolution. The rationale for this design is twofold: (i) longer sequences typically contain more gesture phases, requiring larger windows to capture complete periodic cycles, while shorter sequences represent more compact gestures requiring finer temporal localization; (ii) the divisor α controls the trade-off between frequency resolution (larger windows provide better resolution but reduce temporal localization) and temporal precision (smaller windows track transient dynamics but suffer from spectral leakage). At $\alpha = 4$, the window captures approximately one-quarter of the sequence, providing sufficient context for local frequency analysis while preserving responsiveness to dynamic transitions characteristic of sign language articulation.

A local segment is defined for each timestep t in the sequence, where the segment starts at S_t and ends at E_t. The start and end indices of the segment are defined as follows:

$$S_t = \max(1, t - \lfloor W/2 \rfloor), \tag{28}$$

and

$$E_t = \min(T_n, t + \lfloor W/2 \rfloor). \tag{29}$$

The local segment is used to compute the local FFT for each feature i at timestep t:

$$X_{t,i}^{(n)}[k] = \sum_{\tau=S_t}^{E_t} \tilde{\mathbf{p}}_{\tau,i}^{(n)} e^{-j2\pi(\tau - S_t)k/(E_t - S_t + 1)}. \tag{30}$$

For a small set of coefficients $k = 0, \ldots, C$ (e.g., $C = 4$), the peak magnitude coefficient is selected:

$$k^* = \arg \max_{0 \le k \le C} |X_{t,i}^{(n)}[k]|, \tag{31}$$

The selection of only the peak magnitude coefficient is motivated by the observation that sign language gestures are typically dominated by a single fundamental frequency corresponding to the primary movement rhythm-selecting the peak captures this dominant oscillatory pattern while avoiding noise from secondary harmonics or spectral artifacts.

The corresponding magnitude and normalized phase are defined as:

$$m_{t,i}^{(n)} = |X_{t,i}^{(n)}[k^*]| \tag{32}$$

and

$$\phi_{t,i}^{(n)} = \frac{\arg(X_{t,i}^{(n)}[k^*])}{\pi}. \tag{33}$$

Therefore, the final feature vector at timestep t is defined as:

$$\bar{\mathbf{p}}_t^{(n)} = [\mathbf{p}_t^{(n)} \mid \mathbf{v}_t^{(n)} \mid \mathbf{a}_t^{(n)} \mid \mathbf{m}_t^{(n)} \mid \boldsymbol{\phi}_t^{(n)}] \in \mathbb{R}^{3L_p+A+2L_p}. \tag{34}$$

Here $\mathbf{p}_t^{(n)} \in \mathbb{R}^{L_p}$ denotes the raw spatial position features at timestep t of sequence n, while $\bar{\mathbf{p}}_t^{(n)} \in \mathbb{R}^{L}$ represents the enhanced feature vector and $\tilde{\mathbf{p}}_t^{(n)}$ represents the windowed input for FFT computation.

The position features ($\mathbf{p}_t^{(n)}$), representing the raw spatial coordinates of the joints, are the foundation of SLR, capturing critical hand shapes, configurations, and the spatial context where signs occur. They are especially important for signs where precise location is a defining trait, such as "Happen", whose unique positioning relative to the body aids in differentiation. The velocity features ($\mathbf{v}_t^{(n)}$), describing the rate of positional change, encode the dynamic aspects of signing by revealing movement direction and speed, which are often consistent within the same sign and thus key for reliable recognition. The acceleration features ($\mathbf{a}_t^{(n)}$), measuring how quickly velocity changes, highlight movement starts, stops, and transitions, capturing emphasis points or abrupt directional changes; their peaks often correspond to decisive moments in sign execution. Finally, the FFT features, which transform movement data into the frequency domain, detect repetitive and rhythmic patterns, subtle oscillations, and movement complexity, proving particularly valuable for identifying signs with repetitive motions or characteristic rhythmic structures common in Libras.

2) DEEP NEURAL NETWORK MODEL

The proposed deep learning model for sign language recognition consists of two main components: a feature extraction model and a classification model. The feature extraction model processes the input video frames to extract spatial and temporal features, while the classification model predicts the sign language gestures based on the extracted features. The feature extraction model is based on the procedure described in section III-B. The spatial features include the 3D coordinates of the hand and pose landmarks, while the temporal features capture motion dynamics through kinematic descriptors (velocity and acceleration) and spectral descriptors (FFT-based features). The classification model is a deep neural network that takes the extracted features as input and outputs the predicted sign language gestures. The model is designed to handle the sequential nature of sign language data, capturing temporal dependencies and variations in signing speed and style.

Fig. 5 illustrates the complete architecture of the proposed approach, which can be understood as a two-stage pipeline. In the first stage (Feature extraction model), raw video frames are processed through MediaPipe's Hand and Pose detection modules to extract spatial landmarks representing hand configurations and upper body positioning. These landmarks are then augmented with kinematic features computed from temporal derivatives: velocity vectors capture the speed and direction of movement, while acceleration vectors identify moments of dynamic change such as gesture initiation and termination. Additionally, the adaptive sliding-window FFT is applied to the positional time series to extract spectral characteristics, including dominant frequency magnitudes and normalized phase information, which encode the rhythmic and periodic patterns inherent to signing. The combined feature vector at each timestep thus integrates positional, kinematic, and spectral information, forming a multi-modal representation. In the second stage (Classification model), these feature sequences are fed into a stacked bidirectional Long Short-Term Memory (BiLSTM) network, which processes the temporal dependencies in both forward and backward directions to capture contextual relationships across frames. The BiLSTM outputs are then passed through a series of Fully Connected (FC) layers with ReLU activation, progressively condensing the high-dimensional temporal representations into a compact feature space. Finally, a softmax layer produces class probabilities over the 20 Libras sign categories.

To define the classification model, we first define the input sequence $\bar{\mathbf{P}}^{(n)} \in \mathbb{R}^{T_n \times L}$, where T_n is the number of timesteps in the n-th sequence and L is the number of features per timestep. The input sequence is processed by a stacked BiLSTM network, followed by a sequence of FC layers, and finally outputs logits for classification. Therefore, given an input sequence of length T_n:

$$\bar{\mathbf{P}}^{(n)} = (\bar{\mathbf{p}}_1^{(n)}, \ldots, \bar{\mathbf{p}}_{T_n}^{(n)}). \tag{35}$$

Define $\phi_{\mathrm{LSTM}}(\bar{\mathbf{P}}^{(n)}; \theta_{\mathrm{LSTM}})$ as the stacked bidirectional LSTM layers applied on the input sequence $\bar{\mathbf{P}}^{(n)}$. Each bidirectional LSTM layer processes $\bar{\mathbf{P}}^{(n)}$ to produce hidden states at each time step:

$$\overrightarrow{\mathbf{h}}_t^{(l)} = \mathrm{LSTM}_{fwd}^{(l)}(\mathbf{h}_t^{(l-1)}), \quad \overleftarrow{\mathbf{h}}_t^{(l)} = \mathrm{LSTM}_{bwd}^{(l)}(\mathbf{h}_t^{(l-1)}), \tag{36}$$

$$\mathbf{h}_t^{(l)} = [\overrightarrow{\mathbf{h}}_t^{(l)}; \overleftarrow{\mathbf{h}}_t^{(l)}], \qquad \mathbf{h}_t^{(0)} = \bar{\mathbf{p}}_t. \tag{37}$$

where l indexes the LSTM layers and $\mathbf{x}_t^{(1)} = \mathbf{x}_t$. $\mathbf{h}_t^{(l)}$ is the hidden state at time step t of layer l. The final output of the LSTM is:

$$\mathbf{h} = \phi_{\mathrm{LSTM}}(\bar{\mathbf{P}}^{(n)}; \theta_{\mathrm{LSTM}}) = (\mathbf{h}_1, \ldots, \mathbf{h}_{T_n}) \in \mathbb{R}^{T_n \times d}, \tag{38}$$

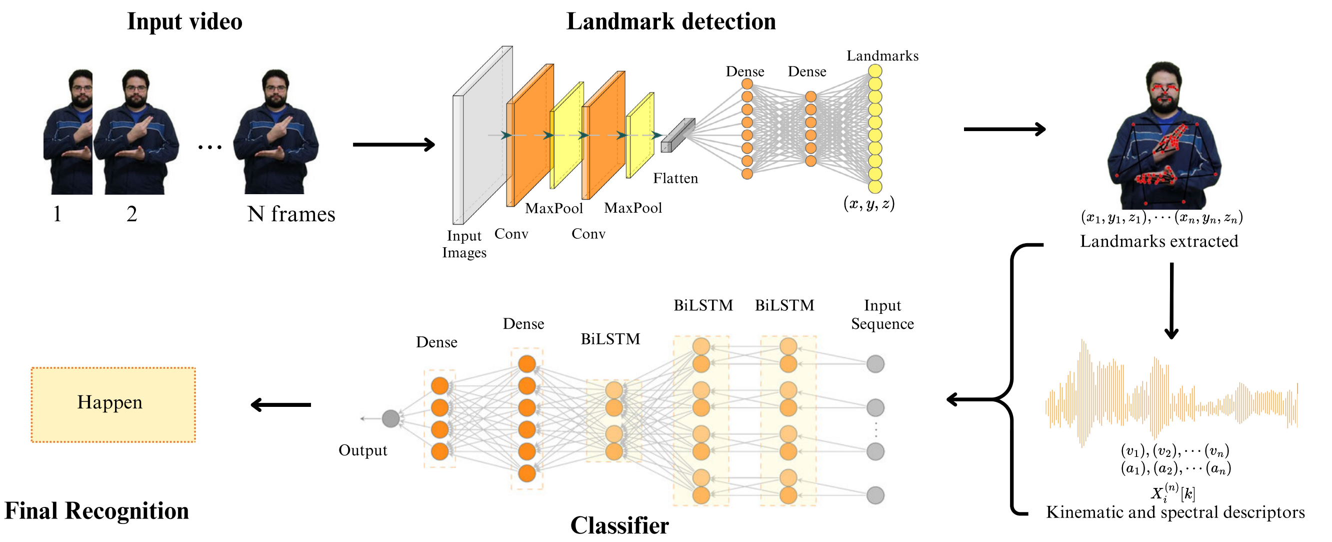


FIGURE 5. Deep learning architecture for sign language recognition, illustrating the input video, which is converted into frame sequences, followed by the feature extraction procedure that combines 3D positional landmarks, kinematic descriptors (velocity v and acceleration a), and spectral descriptors. The extracted features are then fed into a stacked BiLSTM network, followed by fully connected layers, culminating in the final sign classification output.

where d is the hidden state dimension. Note that $\mathbf{h}_t^{(0)} = \bar{\mathbf{p}}_t$ represents the initialization of the first layer's input and $\bar{\mathbf{p}}_t$ represents the feature vector at time step t.

Define $\phi_{\mathrm{FC}}(\mathbf{h}; \theta_{\mathrm{FC}})$ as a sequence of dense layers transforming $\mathbf{h} \in \mathbb{R}^d$ into an intermediate representation $\mathbf{z} \in \mathbb{R}^m$. For each dense layer i:

$$\mathbf{z}^{(i)} = \mathrm{ReLU}(\mathbf{W}^{(i)}\mathbf{z}^{(i-1)} + \mathbf{b}^{(i)}), \quad \mathbf{z}^{(0)} = \mathbf{h}. \tag{39}$$

with $\mathbf{z}^{(0)} := \mathbf{h}$.

The final output of the FC is:

$$\mathbf{z} = \phi_{\mathrm{FC}}(\mathbf{h}; \theta_{\mathrm{FC}}) \in \mathbb{R}^m, \tag{40}$$

and the output logits are computed as:

$$\mathbf{l} = \mathbf{W}_o\mathbf{z} + \mathbf{b}_o, \quad \hat{\mathbf{y}} = \sigma_{\mathrm{softmax}}(\mathbf{l}). \tag{41}$$

where $\mathbf{W}_o \in \mathbb{R}^{C\times m}$ and $\mathbf{b}_o \in \mathbb{R}^C$ are the output layer weights and biases, respectively, and C is the number of classes.

The predicted class probabilities $\hat{\mathbf{y}}$ are:

$$\hat{\mathbf{y}} = \sigma_{\mathrm{softmax}}(\mathbf{l}) = \left[\frac{e^{l_1}}{\sum_j e^{l_j}}, \frac{e^{l_2}}{\sum_j e^{l_j}}, \ldots, \frac{e^{l_{\mathrm{num\_classes}}}}{\sum_j e^{l_j}} \right]. \tag{42}$$

The model is trained to minimize the cross-entropy loss between the predicted class probabilities $\hat{\mathbf{y}}$ and the true labels $\mathbf{y}$, which are one-hot encoded vectors representing the ground truth classes. The loss function over dataset $\{(\bar{\mathbf{P}}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^N$ is defined as follows:

$$\begin{aligned} \mathcal{J}(\theta) &= \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}\big(\mathbf{y}^{(i)}, \hat{\mathbf{y}}^{(i)}\big) \\ &= -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_c^{(i)} \log \hat{y}_c^{(i)}. \end{aligned} \tag{43}$$

The model is designed with three BiLSTM layers, each with 128, 192, and 128 hidden units, respectively, followed by two fully connected layers with 192 and 96 units, respectively, and the ReLU activation function. The final output layer has a softmax activation function to produce class probabilities for the 20 sign language gestures. The model is trained using the Adam optimizer [58] with a learning rate of 0.001 and a batch size of 32.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we present the experimental results of the proposed SLR model. The model was trained and evaluated on the MINDS-Libras dataset. The dataset was split into training and test sets, with 80% of the data used for training and 20% for testing. Cross-validation was performed with K-folds, where the number of folds was set to 10. The model was trained for 200 epochs, and the best model was selected based on the validation accuracy. Methods such as dropout and early stopping were employed to prevent overfitting and improve generalization. The model's performance was evaluated using accuracy, precision, recall, and F1-score metrics.

The models were implemented using Python version 3.10 and the TensorFlow framework. The experiments were conducted on a system equipped with a QEMU Virtual CPU (version 2.5+ (14)) running at 3.312 GHz, an NVIDIA GeForce GTX 1660 Ti GPU, and 30 GiB of RAM. The operating system used was Ubuntu 22.04.3 LTS (x86_64).

For the adaptive sliding-window FFT, the parameters were set as follows: $\alpha = 4$, $W_{\min} = 16$ frames, and $W_{\max} = 64$ frames. The sampling rate is $f_s = 30$ Hz, corresponding to the video frame rate of the MINDS-Libras dataset. The analyzed frequency range is limited to $f_{\min} = 0.5$ Hz and $f_{\max} = 5.0$ Hz, encompassing the physiologically relevant motion spectrum for human signing. The choice of $\alpha = 4$ ensures that the adaptive window captures approximately one-quarter of the gesture sequence, providing sufficient temporal context for local frequency analysis while preserving responsiveness to transient motion patterns.

A Hanning window function $w(t)$ is applied. The number of top FFT coefficients selected is $n_{\text{coeffs}} = 8$, chosen to balance feature dimensionality with computational efficiency while retaining the dominant frequency components. These parameter values were determined empirically through cross-validation experiments that balanced frequency resolution, temporal localization, and computational efficiency.

A. PERFORMANCE ANALYSIS

The performance of the proposed SLR model is evaluated using a confusion matrix and class-wise performance metrics. Fig. 6 shows the confusion matrix of the model across 20 classes. The model demonstrates good accuracy for testing data, as most of the values are concentrated along the diagonal, indicating that the predicted labels match the true labels in the majority of cases. For example, the class "America" is classified correctly, with all instances correctly predicted. Some minor misclassifications are observed in classes such as "Bathroom," "Meet," and "Corner," where an instance is incorrectly predicted as other classes. Other classes, such as "Yellow," "Noise," "Mirror," and "Five," show a few misclassifications to other classes. For example, "Yellow" is misclassified as "Candy" three times, and "Noise" is misclassified as "Enjoy", "Candy" and "Bank".

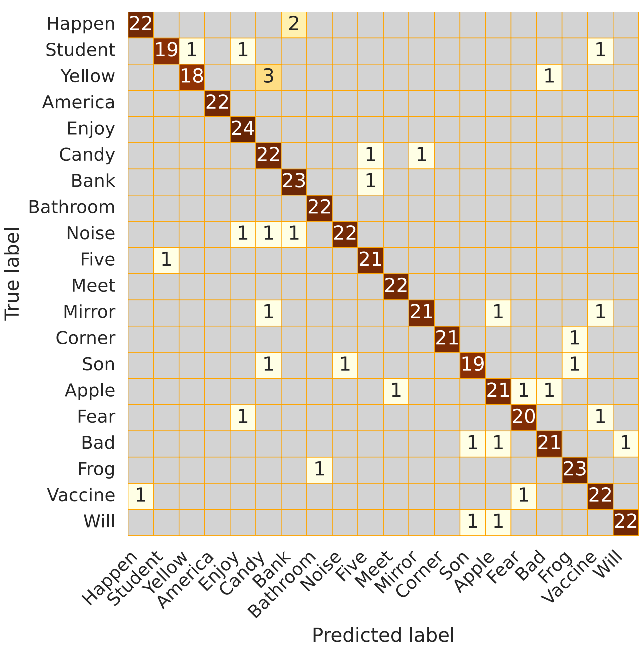


FIGURE 6. Confusion matrix illustrating the classification performance of the model on the test dataset across 20 classes. The matrix shows the true versus predicted labels. Diagonal values represent correct classifications, while off-diagonal values indicate misclassifications. The most confused pairs (e.g., 'Yellow"-"Candy", "Happen"-"Bank") share similar spatial locations, hand trajectories, or movement orientations.

The confusion analysis reveals that most errors occur between signs with overlapping spatial or kinematic characteristics. The most frequent confusion, between "Yellow" and "Candy" (3 instances), arises maybe because both signs are articulated near the face and involve short-hand trajectories, particularly when handshape details, such as an open hand versus an extended finger, are not well captured. Additional mix-ups, such as "Noise" being misclassified as "Candy", "Bank", or "Enjoy", suggest that the model occasionally struggles with signs produced in the central chest or face region, where hand positions often overlap. The confusion between "Happen" and "Bank" (2 instances) likely stems from similar forward motion directions and hand orientations. Other isolated errors, including "Apple" misclassified as "Fear" or "Bad", and "Bad" as "Son" or "Apple", reflect normal intra-signer variability, lighting differences, or temporary occlusions. Overall, the classifier demonstrates high discriminative ability, with misclassifications concentrated among signs that exhibit small or subtle motions, and differ mainly by handshape configuration, features that may be underrepresented when relying solely on skeletal or keypoint-based representations.

Table 3 presents detailed class-wise performance metrics for the proposed approach. Overall, the model demonstrates high accuracy across all classes, with values ranging from 0.9827 to 1.0000. Classes like "America" achieved good scores in accuracy, precision, recall, and F1-score, indicating good recognition.

TABLE 3. Performance test metrics per class for sign language recognition.

Class	Accuracy	Precision	Recall	F1-Score
America	1.0000	1.0000	1.0000	1.0000
Bathroom	0.9978	0.9565	1.0000	0.9778
Meet	0.9978	0.9565	1.0000	0.9778
Corner	0.9978	1.0000	0.9545	0.9767
Enjoy	0.9935	0.8889	1.0000	0.9412
Frog	0.9935	0.9200	0.9583	0.9388
Happen	0.9935	0.9565	0.9167	0.9362
Will	0.9935	0.9565	0.9167	0.9362
Five	0.9935	0.9130	0.9545	0.9333
Bank	0.9914	0.8846	0.9583	0.9200
Noise	0.9914	0.9565	0.8800	0.9167
Mirror	0.9914	0.9545	0.8750	0.9130
Fear	0.9914	0.9091	0.9091	0.9091
Student	0.9914	0.9500	0.8636	0.9048
Vaccine	0.9892	0.8800	0.9167	0.8980
Bad	0.9892	0.9130	0.8750	0.8936
Son	0.9892	0.9048	0.8636	0.8837
Yellow	0.9892	0.9474	0.8182	0.8780
Apple	0.9870	0.8750	0.8750	0.8750
Candy	0.9827	0.7857	0.9167	0.8462

Most classes achieve precision and recall above 0.9, yielding strong F1-scores and indicating a balanced performance. Slight drops appear for "Candy" and "Apple", reflecting occasional misclassifications. The sign "Candy" is often confused with other signs such as "Yellow," "Noise," "Mirror," and "Son," due to similarities in the articulation region and hand configuration. Like "Mirror," it is performed with an open hand and the palm facing the face, making distinction difficult when the motion direction, circular in "Candy" and lateral in "Mirror" is not clearly captured. More subtle confusions also occur with "Yellow","Son," and "Noise," as all involve movements near the face, reducing spatial separability between classes. These morphological and spatial overlaps explain potential ambiguities detected by the model, especially under capture conditions where fine differences in trajectory and orientation are less evident. Conversely, a few true "Apple" samples are mislabeled as "Meet," "Bad," or "Fear," signs may be due to overlapping spatial and dynamic characteristics. The sign "Apple" is performed near the mouth with a "C"-shaped hand and forearm motion; it shares a facial articulation region with "Bad" and a short, unidirectional movement with "Fear". Under capture conditions, the forearm rotation may resemble the partial finger movement in "Bad" or the repetitive hand action on the chest in "Fear" if the exact hand position is not clearly detected. Additionally, "Apple" can be confused with "Meet" when the system fails to misinterprets the approaching linear movement, particularly in initial frames where intermediate hand positions appear similar, both are performed in nearby regions, one at the chin and the other in front of the mouth. These coincidences in execution region and limited motion amplitude contribute to interclass ambiguities in video-based recognition models.

Classes like "Bathroom," "Meet," and "Corner" show minor differences between precision and recall, indicating the model may be slightly more prone to either false positives or false negatives, depending on the class. Notably, some classes ("Yellow" and "Student") show a noticeable drop in recall compared to precision, suggesting these signs are sometimes confused with others.

Signs that share similar articulation regions present notable challenges for the recognition model. For example, "Yellow" and "Candy" both involve movements near the face with short trajectories and minimal arm displacement, making them visually alike in frontal views, especially when finger extension or palm orientation is not clearly captured. Likewise, signs such as "Happen" and "Bank" share overlapping spatial positions and comparable hand orientations, which can lead to confusion when kinematic cues dominate the representation. These similarities cause the model to rely on coarse motion or position features, while fine-grained differences in handshape or contact point, such as the extended index finger in "Yellow" versus the open hand in "Candy", may be underrepresented. This limitation is inherent to skeletal or keypoint-based inputs, which capture joint trajectories but not detailed hand geometry or surface contact. As a result, signs distinguished primarily by subtle finger configurations or orientation nuances tend to produce higher misclassification rates, highlighting the need for richer handshape modeling or multimodal fusion with appearance-based features.

Fig. 7 shows the contribution of each feature group to the classification of individual signs. Warmer colors denote higher importance. Position features dominate most signs, reflecting the central role of spatial configuration in sign discrimination. However, other features gain prominence for particular classes: Magnitude and Phase exhibit notably high importance for "Son", "Fear", "Will", and "Candy", suggesting that temporal and oscillatory motion patterns are key to their recognition. Velocity and Acceleration show lower overall contributions but remain discriminative for certain dynamic signs such as "Happen" and "Corner". The wide importance ranges across all feature groups confirm that the model leverages different kinematic and spectral cues depending on the specific articulatory characteristics of each sign, rather than relying on a uniform feature subset.

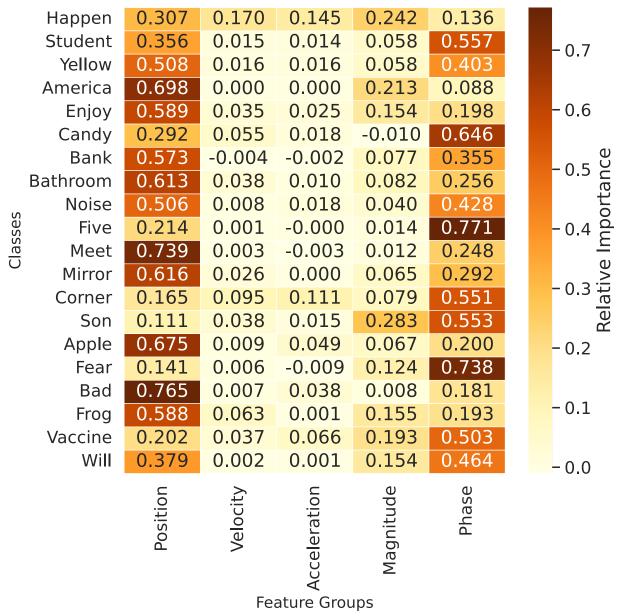
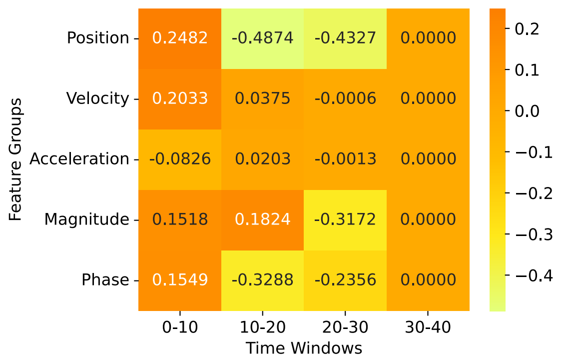


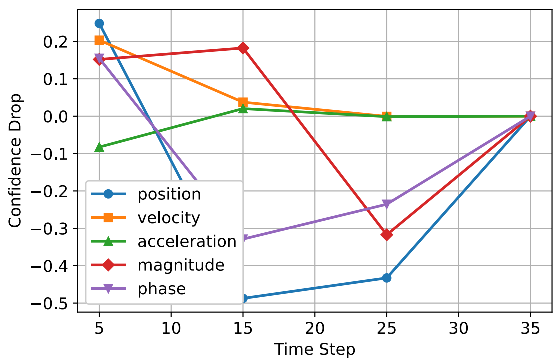
FIGURE 7. Average feature importance per class. Different sign classes rely on distinct feature groups.

Fig. 8 shows the importance of different features over time, for instance 0, which belongs to class "Fear". Early sequence segments (0–15 frames) show positive contributions from velocity and magnitude, indicating that these motion cues are most discriminative for the correct class at the start. The position has a minimal negative impact initially, but becomes detrimental between frames 15 and 25. Additionally, the magnitude shifts from positive to strongly negative after frame 20, suggesting that later movements introduce confusion with other classes. Phase and acceleration exhibit localized effects, phase alternates between negative and positive spikes, and acceleration provides a small positive influence around frames 10–15.

The temporal dynamics revealed in Fig. 8 offer critical insights into how sign language recognition should model the sequential structure of signing gestures. The early dominance of velocity and magnitude features, followed by their decline or reversal in later frames, demonstrates



(a)



(b)

FIGURE 8. **Temporal feature importance for instance 0 of class "fear" over time: (a) average importance score for that feature in the respective time window and (b) feature importance over time.**

that different phases of a sign carry different discriminative information, a finding that challenges approaches treating sign sequences as temporally uniform. Specifically, for "Fear," the initial rapid movement (captured by velocity and magnitude) establishes the sign's identity, while later frames may introduce ambiguity due to transition movements toward the next sign or rest position. The negative contribution of magnitude after frame 20 is particularly revealing: it suggests that prolonged or exaggerated movements in the final phase actually decrease classification confidence, possibly because they deviate from the canonical execution pattern or introduce characteristics similar to other signs.

Fig. 9 provides a comprehensive visualization of feature importance across all test instances, revealing distinct patterns in how different feature modalities contribute to model predictions. The analysis reveals a clear hierarchy of feature importance. Position features exhibit the highest mean impact, with a wide horizontal spread indicating substantial variability in contribution across different signs. The dominance of position features over kinematic features reflects the fundamental nature of sign language phonology. In Libras, as in other sign languages, handshape and spatial location constitute primary phonological parameters that uniquely identify signs. Different signs are often distinguished by where the hands are positioned relative to the body and face,

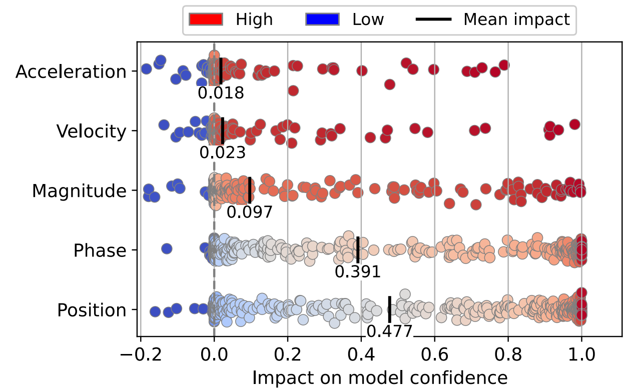


FIGURE 9. **Distribution of feature impacts on model confidence across all test instances, with each point representing a single instance's contribution and colors indicating high (red) or low (blue) importance.**

rather than solely by how they move. The dense concentration of red points demonstrates that spatial hand configurations consistently provide strong discriminative information. Phase features rank second in importance (mean impact of 0.391), showing a similar pattern where higher phase values (red points) predominantly contribute positively to predictions. This underscores the critical role of temporal synchronization patterns in distinguishing between signs with similar spatial characteristics.

In contrast, magnitude features display moderate influence (mean impact of 0.097) with a more balanced distribution across positive and negative impact values. The color gradient reveals that both high and low magnitude values can contribute to correct predictions, suggesting that this feature is sign-dependent; some signs are characterized by large movements, while others require subtle gestures. Velocity and acceleration features show the smallest average impacts, with most points clustered near zero. However, the presence of red points at both positive and negative extremes indicates that while these kinematic features have limited influence on average, they can be decisive for specific signs involving rapid or dynamic movements.

B. ABLATION STUDY ON FEATURE COMPONENTS

To assess the contribution of each feature component to the overall model performance, we conducted an ablation study by systematically removing one feature group at a time and evaluating the resulting model. The results are summarized in Table 4 and Table 5.

The baseline model, relying solely on positional information, achieves an accuracy of 0.88 ± 0.03, confirming that spatial hand configurations already provide substantial discriminative power. When kinematic features are added, a modest yet consistent gain is observed (+1% in both accuracy and F1-score), highlighting the benefit of capturing motion dynamics. The inclusion of spectral descriptors derived from the FFT further enhances the results, reaching 0.92 ± 0.03 accuracy and F1-score. This improvement indicates that frequency-domain cues contribute complementary information beyond positional and kinematic domains, leading to more robust temporal representations. Although total training and inference times increase moderately with model complexity, the accuracy gains justify the additional computational cost, confirming the effectiveness of the proposed multimodal feature combination.

TABLE 4. Feature ablation results showing the contribution of kinematic and spectral components.

Configuration	Parameters	Total Training Time (s)	Accuracy	F1-Score
Baseline	1,601,716	956.04	0.88 ± 0.03	0.88
Position + Kinematic	1,964,212	1029.75	0.89 ± 0.03	0.89
Position + Kinematic + FFT	2,326,708	1226.35	0.92 ± 0.03	0.92

Table 5 shows that adding kinematic and spectral features slightly increases inference latency while enriching motion representation. Feature extraction time rises notably from 0.04 ms/sample to 28.99 ms/sample with spectral descriptors, reflecting the higher cost of frequency analysis. Nonetheless, per-video inference times increase only moderately across configurations.

TABLE 5. Feature extraction and inference timing performance.

Configuration	Feature Extraction (ms/sample)	Mean Total Inference Time (s)	Single Video Inference (ms)
Baseline	0.04	2.5994 ± 0.5734	222.89 ± 26.92
Kinematic Only	0.27	3.2326 ± 0.7033	258.34 ± 47.20
Position + Kinematic + FFT	28.99	3.4564 ± 1.0320	281.46 ± 68.11

In order to further illustrate the effectiveness of the proposed approach, we present a case study of the sign "Happen" in Fig. 10. The sign is characterized by a specific hand movement pattern, where both hands move in a coordinated manner to convey the meaning of the sign. The proposed model effectively captures this pattern by utilizing the spatial hand landmark trajectories, kinematic profiles (velocity and acceleration), and frequency-domain features (FFT magnitude and phase). The proposed model predicted the class "Happen" with a confidence of 9.999999×10^{-1}, followed by "Will" (5.99×10^{-8}) and "Apple" (3.16×10^{-8}). In contrast, the BiLSTM model without kinematic profiles and frequency-domain features predicted "Happen" with a lower confidence of 9.951213×10^{-1}, followed by "America" (1.31677×10^{-3}) and "Vaccine" (1.18654×10^{-3}). These results highlight how the inclusion of kinematic and spectral features contributes to achieving more confident predictions.

The landmark trajectories show a coordinated bilateral movement, where both hands follow similar spatial paths, and the speed and acceleration profiles highlight distinct peaks corresponding to the stroke phase of the sign. The frequency-domain representation further enhances discrimination: the FFT magnitude indicates that most motion energy is concentrated in low-to-mid temporal frequencies, reflecting smooth and deliberate movements, while the phase information encodes the temporal alignment between motion components.

C. LEAVE-ONE-SIGNER-OUT CROSS-VALIDATION

To further assess the model's generalization capabilities across different signers, we conducted a leave-one-signer-out cross-validation (LOSOCV) evaluation. In this approach, the model is trained on data from all but one signer and tested on the held-out signer's data. This process is repeated for each signer in the dataset, ensuring that the model's performance is evaluated on unseen individuals [12].

The distribution of signers is as follows: signer 01 (8.5%), signer 02 (8.6%), signer 03 (7.2%), signer 04 (8.3%), signer 05 (8.5%), signer 06 (8.9%), signer 07 (9.1%), signer 08 (8.6%), signer 09 (6.6%), signer 10 (8.6%), signer 11 (8.4%), and signer 12 (8.7%).

Table 6 summarizes the performance metrics for each signer using LOSOCV. The results indicate variability in performance across different signers, with accuracies ranging from 0.1567 (signer 03) to 0.9672 (signer 09). The mean accuracy across all signers is 0.7701, with a standard deviation of 0.2214, reflecting the challenges of cross-signer generalization in SLR. Moreover, the overall LOSOCV reveals a noticeable drop in performance compared to the signer-dependent scenario. While the signer-dependent model achieved an accuracy of approximately 92%, the mean accuracy across unseen signers decreased to $77.0\% \pm 22.1\%$. This decline indicates that the model's generalization capability is limited when exposed to previously unseen individuals, highlighting the strong influence of signer-specific motion and appearance patterns in the data.

TABLE 6. Performance metrics per test signer using LOSOCV.

Signer	Accuracy	Precision	Recall	F1-Score
1	0.9241	0.9369	0.9241	0.9250
2	0.8742	0.9041	0.8742	0.8703
3	0.1567	0.1684	0.1567	0.1030
4	0.6340	0.6600	0.6340	0.6140
5	0.8981	0.9069	0.8981	0.8947
6	0.8364	0.8261	0.8364	0.8192
7	0.9643	0.9683	0.9643	0.9633
8	0.8938	0.9176	0.8938	0.8901
9	0.9672	0.9726	0.9672	0.9670
10	0.7610	0.7797	0.7610	0.7512
11	0.5548	0.7220	0.5548	0.5844
12	0.7764	0.8220	0.7764	0.7676
Mean $\pm$ Std	0.7701 ± 0.2214	0.7987 ± 0.2119	0.7701 ± 0.2214	0.7625 ± 0.2320

D. COMPARISON WITH OTHER MODELS

We compared the proposed method with state-of-the-art approaches, like De Castro et al. [12], Passos et al. [11],

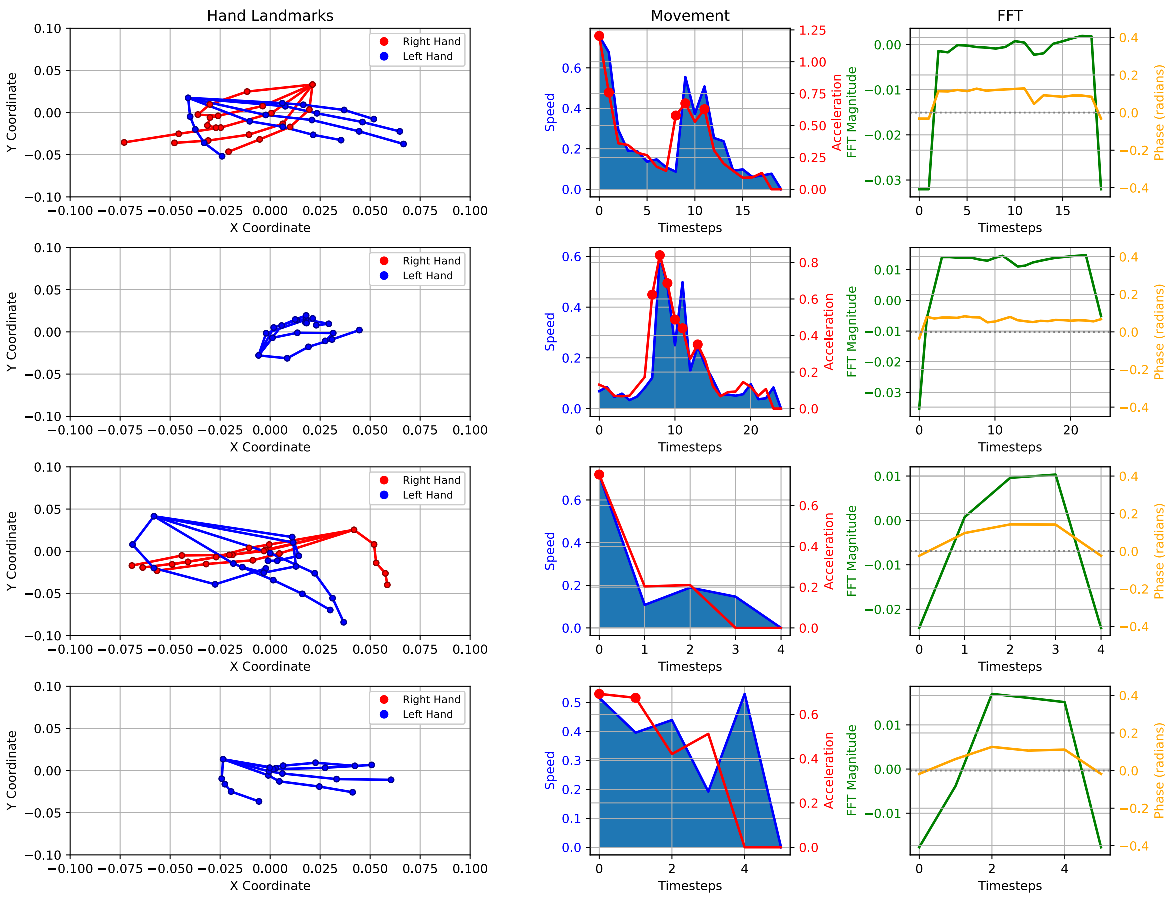


FIGURE 10. **Temporal pattern analysis for the sign "Happen", showing spatial hand landmark trajectories (left), kinematic profiles of speed and acceleration (center), and frequency-domain features via FFT magnitude and phase (right). The average velocity magnitude is 0.2411 ± 0.0724, and the average acceleration magnitude is 0.2957 ± 0.0701. Spectral analysis yields an average FFT magnitude of 0.0001 ± 0.0060. Note that the speed unit is a normalized coordinate per frame.**

and Arcanjo et al. [45], as well as baseline deep learning models such as LSTM, BiLSTM, and CNN. The models were evaluated on the MINDS-Libras dataset in the same conditions, and the results are summarized in Table 7.

The proposed model achieved the highest accuracy, outperforming all competing baselines. This represents a slight but meaningful improvement over De Castro et al. [12], whose model reached 0.91 ± 0.07, and a more substantial margin of 7–8 percentage points over earlier approaches such as Passos et al. [11] and Arcanjo et al. [45]. When compared with standard neural baselines (LSTM, BiLSTM, and CNN), the proposed approach demonstrates a clear advantage in accuracy, highlighting the effectiveness of its feature representation and temporal modeling strategies in capturing the spatiotemporal dynamics of sign gestures.

The performance gains over existing methods highlight key insights into effective SLR. The 1% improvement over De Castro et al. [12], though numerically small, is statistically significant given the high baseline (91%) and reflects an 11% error reduction (from 9% to 8%). Our 3-point advantage over CNN-only models confirms that temporal modeling is essential for dynamic sign recognition, while the 8% gap between LSTM (84%) and BiLSTM (89%) underscores the importance of bidirectional temporal context. Moreover, the 3% improvement over the BiLSTM baseline using only positional features validates the benefit of incorporating kinematic and spectral descriptors.

TABLE 7. **Comparison of state-of-the-art and proposed methods on the MINDS-Libras dataset.**

Method	Accuracy	F1-Score
Castro et al. [12]	0.91 ± 0.07	0.90
Passos et al. [11]	0.85 ± 0.02	–
Arcanjo et al. [45]	0.86 ± 0.08	0.87
Proposed	0.92 ± 0.03	0.92
LSTM	0.84 ± 0.04	0.84
BiLSTM	0.89 ± 0.04	0.89
CNN	0.89 ± 0.04	0.89

Moreover, we also compared the proposed method with the work of Rezende et al. [35], which is the original implementation of the MINDS-Libras dataset, and the work of Arcanjo et al. [45], which is one of the most recent state-of-the-art methods for Libras SLR. The results are summarized in Table 8. To ensure a fair comparison, we used the same separation technique proposed in MINDS, with a random split of 75% of the dataset for training and 25% for testing.

However, it is important to note that this technique can lead to overfitting, as discussed in [12] and [35], which highlights the potential pitfalls of such a split in sign language recognition tasks. We used this split solely for comparison purposes with the mentioned works.

TABLE 8. Comparison of methods for recognizing Libras signs using a 75% training and 25% testing split on the MINDS-Libras dataset. Note: These results may reflect dataset-specific overfitting and should be interpreted cautiously.

Method	Accuracy	F1-Score
Rezende et al. [35]	0.93 ± 0.02	0.93
Arcanjo et al. [45]	0.98 ± 0.01	0.98
Proposed	0.99 ± 0.02	0.98

The proposed method outperforms both Rezende et al. and Arcanjo et al., achieving an accuracy of 0.99 and an F1-score of 0.98. This improvement can be attributed to the proposed feature extraction method, which combines spatial hand landmark trajectories, kinematic profiles, and frequency-domain features. However, as noted above, these comparisons should be interpreted with caution, given the evaluation methodology's limitations. Our confidence in the proposed approach rests primarily on the 10-fold cross-validation results (92% accuracy), which surpass the state-of-the-art method.

The high accuracy (99%) observed with this split likely reflects favorable signer distribution rather than true generalization capability, as evidenced by the substantial performance gap compared to our more robust 10-fold evaluation. Therefore, the 75/25 results are included solely for methodological completeness and direct comparison with prior literature that employed this evaluation protocol.

E. REAL-WORLD TESTS

To evaluate the model's performance in real-world scenarios, we conducted tests with controlled (indoor) and uncontrolled (outdoor) environments. The controlled environment ensured consistent lighting and camera positioning, while the uncontrolled environment introduced variability in lighting and background conditions. The recorded data was then processed using the proposed model to assess its ability to recognize and classify the signs accurately.

In each environment, 20 videos were recorded using five different signers. Each video was 5 seconds long and captured at 60 frames per second. The signers had varying levels of prior knowledge of Libras, including beginners and intermediates, ensuring that the signs were performed with reasonable accuracy while still reflecting natural variability in real-world conditions. During processing, the average time for feature extraction was 0.148 seconds, while the proposed model required an average of 0.198 seconds for inference. The real-world experiments were performed on a system with an Intel® Core™ i5-9300H processor (2.40 GHz), an NVIDIA GeForce GTX 1650 GPU, 12 GB of RAM, running Windows 10.

Table 9 presents the classification probabilities of the model's predictions on the recorded real-world data. The results demonstrate high confidence in both outdoor and indoor settings, with most classes achieving probabilities close to 1. In the outdoor data, the model showed lower confidence for the "America" (0.8085) and "Meet" (0.9326) signs. In the indoor data, classes such as "Apple," "Bathroom," "Five," and "Yellow" were consistently predicted with maximum confidence in both environments. These results indicate that, despite a few environment-dependent variations in confidence, the model is generally robust and capable of accurately recognizing Libras signs.

TABLE 9. Classification probabilities for outdoor and indoor situations.

Class	Outdoor probability	Indoor probability
America	0.8085	1.0000
Apple	1.0000	1.0000
Bad	1.0000	0.9975
Bank	0.9876	1.0000
Bathroom	1.0000	1.0000
Corner	0.9884	0.9474
Enjoy	0.9747	0.9999
Fear	1.0000	0.9395
Five	1.0000	1.0000
Frog	0.9998	1.0000
Happen	0.9853	0.9969
Meet	0.9326	1.0000
Mirror	1.0000	0.9998
Noise	1.000	1.000
Student	1.000	0.9996
Vaccine	1.0000	0.9966
Will	0.9999	1.0000
Yellow	1.0000	1.0000
Average	**0.9820**	**0.9932**

The real-world evaluation highlights the practical potential of the proposed approach. The average probabilities were 0.9932 (indoor) and 0.9820 (outdoor), showing high reliability and only a minor performance gap despite outdoor challenges such as lighting variability, background complexity, and occlusions. The FFT-based spectral features contribute to this robustness by capturing motion rhythms that remain stable under varying illumination. Signs with lower outdoor confidence, such as "America" and "Meet", likely involve hand configurations sensitive to lighting or requiring precise fingertip detection. In contrast, signs like "Apple," "Bathroom," "Five," and "Yellow" achieved good recognition in both settings, indicating distinctive spatial-temporal-spectral signatures. The overall processing time (0.346 s total) supports near real-time operation at about 3 fps, with potential for further optimization toward full real-time performance.

V. CONCLUSION

Deep learning techniques have shown great potential in sign language recognition tasks, particularly when combined with effective feature extraction methods. In this work, we proposed a novel approach that integrates complementary information sources to capture the full complexity of sign language gestures. Unlike previous approaches that rely primarily on spatial hand configurations, the proposed method combines: (i) 3D hand and pose landmarks from MediaPipe for spatial geometric information, (ii) kinematic descriptors (velocity and acceleration) to encode motion dynamics, and (iii) adaptive sliding-window FFT-based spectral features to capture rhythmic patterns and temporal frequencies. Moreover, we proposed a two-stage architecture with feature extraction followed by temporal classification, which demonstrates effectiveness in learning hierarchical representations of signing patterns. The model was trained and evaluated on the MINDS-Libras dataset, achieving state-of-the-art performance with an accuracy of 0.92 and an F1-score of 0.92, surpassing previous methods. The results demonstrate the effectiveness of the proposed feature representation and the deep learning architecture in capturing the complex dynamics of sign language gestures. The model's ability to leverage both spatial and temporal information, along with kinematic and spectral features, allows it to achieve high accuracy and robustness in recognizing sign language gestures. However, it is important to note that the model's performance may vary depending on the specific characteristics of the dataset and the signing styles present in the training data. The proposed method shows promise for real-world applications, such as assistive technologies for the deaf and hard-of-hearing communities, where accurate sign language recognition can facilitate communication and accessibility.

Despite the promising results, the proposed method has some limitations that warrant consideration. First, the computational complexity of the approach is substantial: the feature extraction pipeline involving MediaPipe landmark detection, kinematic computations, and adaptive FFT analysis requires significant processing time per frame, with an average of 0.148 seconds for feature extraction and 0.198 seconds for model inference, which may limit real-time applications on resource-constrained devices. Second, training the stacked BiLSTM architecture with multiple layers demands considerable memory resources and computational power, making it challenging to deploy on edge devices without model compression or optimization techniques. Finally, the method relies heavily on accurate landmark detection, which can degrade in challenging conditions such as poor lighting, occlusions, or non-frontal viewing angles, potentially affecting overall recognition performance.

Additional limitations stem from the dataset and experimental scope. The MINDS-Libras dataset vocabulary is limited to 20 signs, reflecting the current availability of publicly accessible Libras datasets rather than a methodological constraint. Despite this limitation, the present work demonstrates how multi-modal feature integration can improve recognition even within a restricted vocabulary. Future research should focus on expanding the dataset to include a larger and more diverse set of signs to assess scalability.

The field of sign language recognition is rapidly evolving, with deep learning techniques playing a crucial role in advancing the state of the art. The proposed method contributes to this progress by providing a comprehensive feature representation that captures the intricacies of sign language gestures. Future work may explore further enhancements, such as incorporating additional data modalities, improving model interpretability, and expanding the dataset to include more diverse signing styles and contexts, which can be done by the application of Generative Adversarial Networks (GANs) to generate synthetic data. Additionally, the model's performance in real-world scenarios can be further evaluated by testing it in various environments and with different signing styles. Overall, the proposed approach represents a significant step forward in the field of sign language recognition, with the potential to improve communication accessibility for the deaf and hard-of-hearing communities.

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